

SOLAR RACER

2026 & 2027 OFFICIAL RATING FORM

MIDDLE SCHOOL

Judges: Using minimal (1-4 points), adequate (5-8 points), or exemplary (9-10 points) performance levels as a guideline in the rating form, record the scores earned for the event criteria in the column spaces to the right. The X1 or X2 notation in the criteria column is a multiplier factor for determining the points earned. (Example: an "adequate" score of 7 for an X1 criterion = 7 points; an "adequate" score of 7 for an X2 criterion = 14 points.) A score of zero (0) is acceptable if the minimal performance for any criterion is not met.

Go/No Go Specifications

- Before judging the entry, ensure that the items below are present; indicate presence with a check mark in the box.
- If an item is missing, leave the box next to the item blank and place a check mark in the box labeled ENTRY NOT EVALUATED.
- If a check mark is placed in the ENTRY NOT EVALUATED box, the entry is not to be judged.

- ☐ TIER 1 – Solar Car Model Present with student made attachment device
- ☐ TIER 2– Showcase display and QR Code with operational video link
- ☐ ENTRY NOT EVALUATED

TIER 1 – TIME TRIALS		
Fastest Duration Time	Seconds	
TIER 1 – TIME TRIALS TOTAL (Seconds)		

Rules violations (a deduction of 20% of the total combined time trials for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	

TIER 1 – SUBTOTAL (Seconds)	
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TIER 2 – SHOWCASE DISPLAY AND MODEL (130 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance 1-4 points	Adequate performance 5-8 points	Exemplary performance 9-10 points	
Display Components (X1)	A number of display components are missing.	Most of the display components are included, but the quality is minimal.	The display includes all required components in order; it is neat and properly organized; effort and quality are evident.	
Engineering Design Process (X2)	The Engineering Design Process is not defined or is lacking significant portions; it is messy and demonstrates lack of effort.	The Engineering Design Process is acceptable, with most information included.	The Engineering Design Process is complete and accurate; the information is neat and orderly; a great deal of effort is evident.	
Design Drawings (X1)	Some drawings are missing and/or drawings are of poor quality.	Drawings are acceptable; all required views are shown.	Drawings are accurate and complete; all required views are present; rough sketches are included.	

TIER 2 – SHOWCASE DISPLAY AND MODEL (130 points) – continued					
Design Details/ Components List (X1)	Several details of the model, such as model size, wheel size, and gear ratio are missing and/or are poor; the components list is very limited.	Most details of the model, such as model size, wheel size, and gear ratio are included; most components are included.	All details of the model, such as model size, wheel size, and gear ratio are present; all components are included.		
Run Log (X1)	No Run Log created or less than five (5) runs recorded.	Run Log created with multiple runs but little detail related to performance and adjustments.	Run Log included with multiple runs and detailed information about performance, variables and adjustments.		
Videos using QR Code (X1)	The QR code is not included or the videos are not accessible.	The QR code is present and one (1) to two (2) videos are provided as evidence of model car running on solar power alone.	The QR code is present and three (3) videos are provided as evidence of model car running on solar power alone.		
Model Design (X2)	The design of the solar model is poor and shows little effort.	The design of the solar model is adequate but not of exceptional quality.	The design of the solar model exhibits exceptional quality.		
Model Creativity/ Originality (X2)	The solar model car design lacks creativity and originality; little effort is apparent; car is an exact, or nearly an exact replica of purchased kit.	The solar model car design demonstrates an adequate level of creativity and originality; at least one (1) modification has been made to the car.	The solar model car design shows exceptional creativity, originality, artisanship, and engineering.		
Model Construction (X2)	The solar model car lacks quality of construction.	The solar model car demonstrates adequate quality of construction.	The solar model car demonstrates exceptional quality of construction.		
TIER 2 – SHOWCASE DISPLAY AND MODEL SUBTOTAL (130 points)					

TIER 2 – HEAD-TO-HEAD RACE (80 points)							
Points are assigned based on results from double elimination bracket.							
1st	2nd	3rd & 4th	5th – 8th	9th – 16th	17th – 24th		
80 Points	75 Points	70 Points	60 Points	50 Points	40 Points		
SEMIFINAL RACE SUBTOTAL (80 points)							

Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right.	
Indicate the rule violated: _____	

PRELIMINARY TOTAL (210 points)	
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SEMIFINAL CAR BUILDER INTERVIEW (40 points)				Record scores in the column spaces below.
CRITERIA	Minimal performance	Adequate performance	Exemplary performance	
	1-4 points	5-8 points	9-10 points	
Car Builder Interview (X2)	The team shows very limited knowledge of (and has difficulty articulating) how the car was produced or decisions made during the production; the student exhibits a basic understanding of design elements and functionality, and the rationale is inconsistent or absent.	The team demonstrates some knowledge of the vehicle production and has adequate knowledge of some processes or reasoning behind the vehicle design.	The team shows competence and knowledge related to the design and production of the vehicle; the student is able to articulate "reasoning" behind the decisions made.	
Articulation (X2)	Communication of the design process is unclear, unorganized, and or illogical; leadership and/or 21 st century skills are not evident.	Communication of the design process is somewhat logical and clear; leadership and/or 21 st century skills are somewhat evident.	Communication of the design process is clear, concise, and logical; leadership and/or 21 st century skills are clearly evident.	
SEMIFINAL CAR BUILDER INTERVIEW SUBTOTAL (40 points)				
Rules violations (a deduction of 20% of the total possible points for the above sections) must be initialed by the judge, coordinator, and manager of the event. Record the deduction in the space to the right. Indicate the rule violated: _____				
SEMIFINAL TOTAL (40 points)				
To arrive at the FINAL TOTAL score, subtract rules violation points, as necessary. TOTAL (250 points)				

Comments:

I certify these results to be true and accurate to the best of my knowledge.

JUDGE

Printed name: _____ Signature: _____